AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

FIN340 and FIN340-F

Course Title: ASSET PRICING AND CORPORATE FINANCE

Version C

Mid-term Examination

2025-2026 / Fall

3-December-2025

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umber of Exam Pages: 7 ıcluding this cover sheet)	Time Allowed:1.5 hours

Instructions:

Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
This is a **CLOSED** book examination. NO documents of any kind are allowed.
You **MUST** answer the questions using only the answer booklets provided
You **MUST** return the examination paper together with your answer booklets.
Do **NOT** write your answers in pencil except graphs/sketches
Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinati
Students are allowed to use ***non-programmable*** calculators only (if needed).
Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Answer ALL 5 Questions

- **Question 1: (10 marks)**

You are scheduled to receive $55,750 in four years. When you receive it, you will invest it for 8 more years, at 12 percent per year. How much money will you have 12 years from now?

- **Question 2: (10 marks)**

Suppose you are committed to owning a $140,000 Ferrari. You believe your mutual fund can achieve an annual rate of return of 8 percent and you want to buy the car in 7 years. How much must you invest today to fund this purchase assuming the price of the car remains constant?

- **Question 3: (30 marks)**

What is the sustainable growth rate assuming the following ratios are constant? Interpret your result.

Total asset turnover=3.5
Profit Margin= 20%
Equity multiplier= 1
Payout ratio= 55%

- **Question 4: (30 marks)**

Disruption Tech. has annual sales of $500,000 with 35,000 shares of stock outstanding. The firm has a profit margin of 15 percent and a price-sales ratio of 3.0. What is the firm's price-earnings ratio? Interpret the company's price-earnings ratio.

- **Question 5: (20 marks)**

Explain what kind of decisions financial management is responsible for.

End of Questions

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Formulas

Cash Flow Summary

I. The cash flow identity

Cash flow from assets	=	Cash flow to creditors (bondholders)
	+	stockholders (owners)

II. Cash flow from assets

Cash flow from assets	=	Operating cash flow
	–	Net capital spending
	–	Change in net working capital (NWC)

where:

Operating cash flow	=	Earnings before interest and taxes (EBIT)
	+	Depreciation – Taxes
Net capital spending	=	Ending net fixed assets – Beginning net fixed assets
	+	Depreciation
Change in NWC	=	Ending NWC – Beginning NWC

III. Cash flow to creditors (bondholders)

Cash flow to creditors	=	Interest paid – Net new borrowing

IV. Cash flow to stockholders (owners)

Cash flow to stockholders	=	Dividends paid – Net new equity raised

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Common Financial Ratios

I. Short-term solvency, or liquidity, ratios
$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$
$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$
$\text{Cash ratio} = \frac{\text{Cash}}{\text{Current liabilities}}$
$\text{Net working capital to total assets} = \frac{\text{Net working capital}}{\text{Total assets}}$
$\text{Interval measure} = \frac{\text{Current assets}}{\text{Average daily operating costs}}$
II. Long-term solvency, or financial leverage, ratios
$\text{Total debt ratio} = \frac{\text{Total assets} - \text{Total equity}}{\text{Total assets}}$
$\text{Debt-equity ratio} = \text{Total debt/Total equity}$
$\text{Equity multiplier} = \text{Total assets/Total equity}$
$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Total equity}}$
$\text{Times interest earned ratio} = \frac{\text{EBIT}}{\text{Interest}}$
$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest}}$

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III. Asset management, or turnover, ratios
$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$
$\text{Days' sales in inventory} = \frac{\text{365 days}}{\text{Inventory turnover}}$
$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable}}$
$\text{Days' sales in receivables} = \frac{\text{365 days}}{\text{Receivables turnover}}$
$\text{NWC turnover} = \frac{\text{Sales}}{\text{NWC}}$
$\text{Fixed asset turnover} = \frac{\text{Sales}}{\text{Net fixed assets}}$
$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$
IV. Profitability ratios
$\text{Profit margin} = \frac{\text{Net income}}{\text{Sales}}$
$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets}}$
$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Total equity}}$
$\text{ROE} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets*}}{\text{Equity}}$

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V. Market value ratios

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

$$\text{PEG ratio} = \frac{\text{Price-earnings ratio}}{\text{Earnings growth rate (\%)}}$$

$$\text{Price-sales ratio} = \frac{\text{Price per share}}{\text{Sales per share}}$$

$$\text{Market-to-book-ratio} = \frac{\text{Market value per share}}{\text{Book value per share}}$$

$$\text{Tobin's Q ratio} = \frac{\text{Market value of assets}}{\text{Replacement cost of assets}}$$

$$\text{Enterprise value-EBITDA ratio} = \frac{\text{Enterprise value}}{\text{EBITDA}}$$

ROE= Profit Margin X Total Asset Turnover X Equity Multiplier

Summary of Internal and Sustainable Growth Rates

I. Internal Growth Rate

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}$$

where:

ROA = Return on assets= Net income/Total assets
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The internal growth rate is the maximum growth rate that can be achieved with no external financing of any kind.

II. Sustainable Growth Rate

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b}$$

where:

ROE = Return on equity= Net income/Total equity
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The sustainable growth rate is the maximum growth rate that can be achieved with no external equity financing while maintaining a constant debt-equity ratio.

Dividend payout ratio = Cash dividends/Net income

Retention ratio= 1- the dividend payout ratio

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$$EPS = \frac{\text{Net income}}{\text{Shares outstanding}} =$$

Net Working Capital= Current Asset- Current Liabilities

Present value: $PV = FV/(1 + r)^t$

Future Value: $FV = PV*(1 + r)$

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